四 不定积分

解答题

1. 解 (1) $\int x\sec^4(x^2+1)\mathrm{d}x=\frac{1}{2}\int\sec^4(x^2+1)\mathrm{d}(x^2+1)$

$$=\frac{1}{2}\int[\tan^2(x^2+1)+1]\mathrm{d}[\tan(x^2+1)]$$

$$=\frac{1}{6}\tan^3(x^2+1)+\frac{1}{2}\tan(x^2+1)+C.$$

(2) $\int\frac{x}{4+x^4}\mathrm{d}x=\frac{1}{2}\int\frac{\mathrm{d}(x^2)}{2^2+(x^2)^2}=\frac{1}{4}\arctan\frac{x^2}{2}+C.$

(3) $\int\frac{x\,\mathrm{d}x}{\sqrt{4-x^4}}=\frac{1}{2}\int\frac{\mathrm{d}(x^2)}{\sqrt{2^2-(x^2)^2}}=\frac{1}{2}\arcsin\frac{x^2}{2}+C.$

2. 解 (1) $\int\left(1-\frac{1}{x^2}\right)\tan^2\left(x+\frac{1}{x}\right)\mathrm{d}x=\int\left[\sec^2\left(x+\frac{1}{x}\right)-1\right]\mathrm{d}\left(x+\frac{1}{x}\right)$

$$=\tan\left(x+\frac{1}{x}\right)-x-\frac{1}{x}+C.$$

(2) $\int\frac{x^2+1}{x^4+1}\mathrm{d}x=\int\frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}\mathrm{d}x=\int\frac{\mathrm{d}\left(x-\frac{1}{x}\right)}{(\sqrt{2})^2+\left(x-\frac{1}{x}\right)^2}=\frac{1}{\sqrt{2}}\arctan\frac{x-\frac{1}{x}}{\sqrt{2}}+C.$

3. 解 (1) $\int\frac{\mathrm{d}x}{\sqrt{\mathrm{e}^{2x}-1}}=\int\frac{\mathrm{d}x}{\mathrm{e}^x\sqrt{1-(\mathrm{e}^{-x})^2}}=\int\frac{\mathrm{e}^{-x}\mathrm{d}x}{\sqrt{1-(\mathrm{e}^{-x})^2}}$

$$=-\int\frac{\mathrm{d}(\mathrm{e}^{-x})}{\sqrt{1-(\mathrm{e}^{-x})^2}}=-\arcsin\mathrm{e}^{-x}+C.$$

(2) $\int\frac{\mathrm{e}^{3x}+\mathrm{e}^x}{\mathrm{e}^{4x}+\mathrm{e}^{2x}+1}\mathrm{d}x=\int\frac{\mathrm{e}^x+\mathrm{e}^{-x}}{\mathrm{e}^{2x}+\mathrm{e}^{-2x}+1}\mathrm{d}x=\int\frac{\mathrm{d}(\mathrm{e}^x-\mathrm{e}^{-x})}{(\sqrt{3})^2+(\mathrm{e}^x-\mathrm{e}^{-x})^2}$

$$=\frac{1}{\sqrt{3}}\arctan\frac{\mathrm{e}^x-\mathrm{e}^{-x}}{\sqrt{3}}+C.$$

4. 解 (1) $\int\frac{\mathrm{d}x}{\sqrt{x(1-4x)}}=2\int\frac{\mathrm{d}x}{2\sqrt{x}\sqrt{1-4x}}=2\int\frac{\mathrm{d}(\sqrt{x})}{\sqrt{1-4x}}=\int\frac{\mathrm{d}(2\sqrt{x})}{\sqrt{1-(2\sqrt{x})^2}}$

$$=\arcsin(2\sqrt{x})+C.$$

(2) $\int\frac{\cos\sqrt{x}-1}{\sqrt{x}\sin^2\sqrt{x}}\mathrm{d}x=2\int\frac{\cos\sqrt{x}-1}{2\sqrt{x}\sin^2\sqrt{x}}\mathrm{d}x=2\int\frac{\cos\sqrt{x}-1}{\sin^2\sqrt{x}}\mathrm{d}(\sqrt{x})$

$$=2\left[\int\frac{\mathrm{d}(\sin\sqrt{x})}{\sin^2\sqrt{x}}-\int\csc^2\sqrt{x}\,\mathrm{d}(\sqrt{x})\right]=2\left(-\frac{1}{\sin\sqrt{x}}+\cot\sqrt{x}\right)+C.$$

(3) $\int\frac{\arcsin\sqrt{x}}{\sqrt{x(1-x)}}\mathrm{d}x=2\int\frac{\arcsin\sqrt{x}}{2\sqrt{x}\sqrt{1-x}}\mathrm{d}x=2\int\frac{\arcsin\sqrt{x}}{\sqrt{1-(\sqrt{x})^2}}\mathrm{d}(\sqrt{x})$

$$=2\int\arcsin\sqrt{x}\,\mathrm{d}(\arcsin\sqrt{x})=\arcsin^2\sqrt{x}+C.$$

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[注]$\int\frac{1}{\sqrt{x}}\mathrm{d}x=2\int\frac{1}{2\sqrt{x}}\mathrm{d}x=2\sqrt{x}+C$，即$\frac{1}{\sqrt{x}}=\mathrm{d}(2\sqrt{x})$，遇到分母为$\sqrt{x}$形式的题可以考虑这样变换.

5. 解 (1)$\int\frac{\mathrm{d}x}{x\ln x\cdot(\ln^2x+2)}=\int\frac{\mathrm{d}(\ln x)}{\ln x\cdot(\ln^2x+2)}=\int\frac{\ln x\mathrm{d}(\ln x)}{\ln^2x\cdot(\ln^2x+2)}=\frac{1}{2}\int\frac{\mathrm{d}(\ln^2x)}{\ln^2x\cdot(\ln^2x+2)}$

$$=\frac{1}{4}\int\left(\frac{1}{\ln^2x}-\frac{1}{\ln^2x+2}\right)\mathrm{d}(\ln^2x)=\frac{1}{4}\ln\frac{\ln^2x}{\ln^2x+2}+C.$$

(2)$\int\frac{1+\ln x}{(x\ln x)^2}\mathrm{d}x=\int\frac{\mathrm{d}(x\ln x)}{(x\ln x)^2}=-\frac{1}{x\ln x}+C.$

(3)$\int\frac{1+x}{x(x\mathrm{e}^x+1)}\mathrm{d}x=\int\frac{(1+x)\mathrm{e}^x}{x\mathrm{e}^x(x\mathrm{e}^x+1)}\mathrm{d}x=\int\left(\frac{1}{x\mathrm{e}^x}-\frac{1}{x\mathrm{e}^x+1}\right)\mathrm{d}(x\mathrm{e}^x)=\ln\left|\frac{x\mathrm{e}^x}{x\mathrm{e}^x+1}\right|+C.$

6. 解 (1)令$\sqrt{\mathrm{e}^x-1}=t$，$x=\ln(1+t^2)$，则

$$\int\sqrt{\mathrm{e}^x-1}\,\mathrm{d}x=\int t\cdot\frac{2t}{1+t^2}\mathrm{d}t=2\int\left(1-\frac{1}{1+t^2}\right)\mathrm{d}t=2t-2\arctan t+C$$

$$=2\sqrt{\mathrm{e}^x-1}-2\arctan\sqrt{\mathrm{e}^x-1}+C.$$

(2)$\int\frac{\sqrt{x}}{\sqrt{x}+1}\mathrm{d}x=\int\left(1-\frac{1}{\sqrt{x}+1}\right)\mathrm{d}x=x-\int\frac{1}{\sqrt{x}+1}\mathrm{d}x$，令$\sqrt{x}=t$，则

$$\int\frac{1}{\sqrt{x}+1}\mathrm{d}x=\int\frac{2t}{t+1}\mathrm{d}t=2\int\left(1-\frac{1}{t+1}\right)\mathrm{d}t$$

$$=2t-2\ln(t+1)+C=2\sqrt{x}-2\ln(1+\sqrt{x})+C,$$

故$\int\frac{\sqrt{x}}{\sqrt{x}+1}\mathrm{d}x=x-2\sqrt{x}+2\ln(1+\sqrt{x})+C.$

7. 解 (1)令$x=\sec t$，则

$$\int\frac{\mathrm{d}x}{x\sqrt{x^2-1}}=\int\frac{\sec t\tan t}{\sec t\tan t}\mathrm{d}t=\int\mathrm{d}t=t+C=\arccos\frac{1}{x}+C.$$

(2)令$x=\sec t$，则

$$\int\frac{\sqrt{x^2-1}}{x^2}\mathrm{d}x=\int\frac{\tan t}{\sec^2t}\cdot\sec t\tan t\,\mathrm{d}t=\int\frac{\tan^2t}{\sec t}\mathrm{d}t=\int\frac{\sec^2t-1}{\sec t}\mathrm{d}t$$

$$=\int\sec t\,\mathrm{d}t-\int\cos t\,\mathrm{d}t=\ln|\sec t+\tan t|-\sin t+C$$

$$=\ln|x+\sqrt{x^2-1}|-\frac{\sqrt{x^2-1}}{x}+C.$$

(3)令$x=\sin t$，则

$$\int\frac{\mathrm{d}x}{1+\sqrt{1-x^2}}=\int\frac{\cos t}{1+\cos t}\mathrm{d}t=\int\left(1-\frac{1}{1+\cos t}\right)\mathrm{d}t=t-\int\frac{1-\cos t}{1-\cos^2t}\mathrm{d}t=t-\int\frac{1-\cos t}{\sin^2t}\mathrm{d}t$$

$$=t-\int\csc^2t\,\mathrm{d}t+\int\frac{\mathrm{d}(\sin t)}{\sin^2t}=t+\cot t-\frac{1}{\sin t}+C$$

$$=\arcsin x+\frac{\sqrt{1-x^2}}{x}-\frac{1}{x}+C.$$

(4)令$x=\tan t$，则$\int\frac{\mathrm{d}x}{\sqrt{(x^2+1)^3}}=\int\frac{\sec^2t}{\sec^3t}\mathrm{d}t=\int\cos t\,\mathrm{d}t=\sin t+C=\frac{x}{\sqrt{x^2+1}}+C.$

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8. 解 (1)$\int \frac{\mathrm{d}x}{1+\cos x}=\int \frac{\mathrm{d}x}{2\cos^2\frac{x}{2}}=\int \sec^2\frac{x}{2}\mathrm{d}\left(\frac{x}{2}\right)=\tan\frac{x}{2}+C.$

(2)$\int \frac{\sin x}{1+\sin x}\mathrm{d}x=\int\left(1-\frac{1}{1+\sin x}\right)\mathrm{d}x=x-\int \frac{\mathrm{d}\left(x-\frac{\pi}{2}\right)}{1+\cos\left(x-\frac{\pi}{2}\right)}=x-\tan\left(\frac{x}{2}-\frac{\pi}{4}\right)+C.$

(3)$\int \frac{\mathrm{d}x}{\sin x+\sqrt{3}\cos x}=\frac{1}{2}\int \frac{\mathrm{d}x}{\cos\left(x-\frac{\pi}{6}\right)}=\frac{1}{2}\int \sec\left(x-\frac{\pi}{6}\right)\mathrm{d}\left(x-\frac{\pi}{6}\right)$

$=\frac{1}{2}\ln\left|\sec\left(x-\frac{\pi}{6}\right)+\tan\left(x-\frac{\pi}{6}\right)\right|+C.$

(4)$\int \frac{\mathrm{d}x}{\sqrt{2}+\sin x+\cos x}=\frac{1}{\sqrt{2}}\int \frac{\mathrm{d}\left(x-\frac{\pi}{4}\right)}{1+\cos\left(x-\frac{\pi}{4}\right)}=\frac{1}{\sqrt{2}}\tan\left(\frac{x}{2}-\frac{\pi}{8}\right)+C.$

(5)$\int \frac{\mathrm{d}x}{1+2\cos^2 x}=\int \frac{\sec^2 x}{\sec^2 x+2}\mathrm{d}x=\int \frac{\mathrm{d}(\tan x)}{(\sqrt{3})^2+\tan^2 x}=\frac{1}{\sqrt{3}}\arctan\frac{\tan x}{\sqrt{3}}+C.$

(6)$\int \frac{\mathrm{d}x}{1+\sin x+\cos x}=\int \frac{\mathrm{d}x}{2\cos^2\frac{x}{2}+2\sin\frac{x}{2}\cos\frac{x}{2}}=\int \frac{1}{\cos^2\frac{x}{2}+\sin\frac{x}{2}\cos\frac{x}{2}}\mathrm{d}\left(\frac{x}{2}\right)$

$=\int \frac{\sec^2\frac{x}{2}}{1+\tan\frac{x}{2}}\mathrm{d}\left(\frac{x}{2}\right)=\int \frac{\mathrm{d}\left(1+\tan\frac{x}{2}\right)}{1+\tan\frac{x}{2}}=\ln\left|1+\tan\frac{x}{2}\right|+C.$

9. 解 (1)$\int \frac{\sin x+\cos^2 x}{1+\sin x}\mathrm{d}x=\int\left(1-\frac{1}{1+\sin x}+\frac{1-\sin^2 x}{1+\sin x}\right)\mathrm{d}x=\int\left(2-\sin x-\frac{1}{1+\sin x}\right)\mathrm{d}x$

$=2x+\cos x-\tan x+\frac{1}{\cos x}+C.$

(2)$\int \frac{\mathrm{d}x}{\sin x\cos^4 x}=\int \frac{\sin^2 x+\cos^2 x}{\sin x\cos^4 x}\mathrm{d}x=\int \frac{\sin^2 x}{\sin x\cos^4 x}\mathrm{d}x+\int \frac{\cos^2 x}{\sin x\cos^4 x}\mathrm{d}x$

$=\int \frac{\sin x}{\cos^4 x}\mathrm{d}x+\int \frac{1}{\sin x\cos^2 x}\mathrm{d}x=-\int \frac{\mathrm{d}(\cos x)}{\cos^4 x}+\int \frac{\sin^2 x+\cos^2 x}{\sin x\cos^2 x}\mathrm{d}x$

$=\frac{1}{3\cos^3 x}+\int \frac{\sin^2 x}{\sin x\cos^2 x}\mathrm{d}x+\int \frac{\cos^2 x}{\sin x\cos^2 x}\mathrm{d}x$

$=\frac{1}{3\cos^3 x}-\int \frac{\mathrm{d}(\cos x)}{\cos^2 x}+\int \csc x\,\mathrm{d}x$

$=\frac{1}{3\cos^3 x}+\frac{1}{\cos x}+\ln|\csc x-\cot x|+C.$

(3)$\int \frac{\cos^2 x\,\mathrm{d}x}{1+\cos x}=\int \frac{\cos^2 x-1+1}{1+\cos x}\mathrm{d}x=\int\left(\cos x-1+\frac{1}{1+\cos x}\right)\mathrm{d}x$

$=\sin x-x+\tan\frac{x}{2}+C.$

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(4)$\int\frac{1+\cos x}{1+\sin x}\mathrm{d}x=\int\frac{1}{1+\sin x}\mathrm{d}x+\int\frac{\cos x}{1+\sin x}\mathrm{d}x$

$=\int\frac{1}{1+\cos\left(x-\frac{\pi}{2}\right)}\mathrm{d}x+\int\frac{\mathrm{d}(1+\sin x)}{1+\sin x}$

$=\int\frac{1}{2\cos^2\left(\frac{x}{2}-\frac{\pi}{4}\right)}\mathrm{d}x+\ln(1+\sin x)$

$=\int\sec^2\left(\frac{x}{2}-\frac{\pi}{4}\right)\mathrm{d}\left(\frac{x}{2}-\frac{\pi}{4}\right)+\ln(1+\sin x)$

$=\tan\left(\frac{x}{2}-\frac{\pi}{4}\right)+\ln(1+\sin x)+C.$

10. 解 (1) 令$\frac{x+2}{2x^2-x-1}=\frac{x+2}{(x-1)(2x+1)}=\frac{A}{x-1}+\frac{B}{2x+1}.$

由$A(2x+1)+B(x-1)=x+2$,得$\begin{cases}2A+B=1,\\A-B=2,\end{cases}$解得$A=1,B=-1$,则

$$\int\frac{x+2}{2x^2-x-1}\mathrm{d}x=\int\frac{\mathrm{d}x}{x-1}-\int\frac{\mathrm{d}x}{2x+1}=\ln|x-1|-\frac{1}{2}\ln|2x+1|+C.$$

(2)$\int\frac{x-1}{x^2+x+1}\mathrm{d}x=\frac{1}{2}\int\frac{(2x+1)-3}{x^2+x+1}\mathrm{d}x=\frac{1}{2}\int\frac{\mathrm{d}(x^2+x+1)}{x^2+x+1}-\frac{3}{2}\int\frac{\mathrm{d}\left(x+\frac{1}{2}\right)}{\left(\frac{\sqrt{3}}{2}\right)^2+\left(x+\frac{1}{2}\right)^2}$

$=\frac{1}{2}\ln(x^2+x+1)-\sqrt{3}\arctan\frac{2x+1}{\sqrt{3}}+C.$

(3) 令$\frac{2x+1}{x(1+x^2)}=\frac{A}{x}+\frac{Bx+C}{1+x^2}$,则$A(1+x^2)+x(Bx+C)=2x+1$,解得

$$A=1,B=-1,C=2.$$

$$\int\frac{2x+1}{x(1+x^2)}\mathrm{d}x=\int\left(\frac{1}{x}-\frac{x-2}{1+x^2}\right)\mathrm{d}x=\ln|x|-\frac{1}{2}\ln(1+x^2)+2\arctan x+C.$$

(4)$\int\frac{x+1}{(2x-1)^2}\mathrm{d}x=\frac{1}{4}\int\frac{(2x-1)+3}{(2x-1)^2}\mathrm{d}(2x-1)=\frac{1}{4}\int\frac{\mathrm{d}(2x-1)}{2x-1}+\frac{3}{4}\int\frac{\mathrm{d}(2x-1)}{(2x-1)^2}$

$=\frac{1}{4}\ln|2x-1|-\frac{3}{4(2x-1)}+C.$

(5)$\int\frac{\mathrm{d}x}{x(x^4+3)}=\int\frac{x^3\mathrm{d}x}{x^4(x^4+3)}=\frac{1}{4}\int\frac{\mathrm{d}(x^4)}{x^4(x^4+3)}=\frac{1}{12}\left[\int\frac{\mathrm{d}(x^4)}{x^4}-\int\frac{\mathrm{d}(x^4+3)}{x^4+3}\right]$

$=\frac{1}{12}\ln\frac{x^4}{x^4+3}+C.$

11. 解 (1)$\int x\ln^2x\,\mathrm{d}x=\int\ln^2x\,\mathrm{d}\left(\frac{x^2}{2}\right)=\frac{x^2}{2}\ln^2x-\frac{1}{2}\int x^2\cdot2\ln x\cdot\frac{1}{x}\mathrm{d}x$

$=\frac{x^2}{2}\ln^2x-\int\ln x\,\mathrm{d}\left(\frac{x^2}{2}\right)=\frac{x^2}{2}\ln^2x-\frac{x^2}{2}\ln x+\frac{x^2}{4}+C.$

(2)$\int x^2\ln(1+x^2)\mathrm{d}x=\int\ln(1+x^2)\mathrm{d}\left(\frac{x^3}{3}\right)=\frac{x^3}{3}\ln(1+x^2)-\frac{2}{3}\int\frac{x^4}{1+x^2}\mathrm{d}x$

$=\frac{x^3}{3}\ln(1+x^2)-\frac{2}{3}\int\left(x^2-1+\frac{1}{1+x^2}\right)\mathrm{d}x$

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$$=\frac{x^3}{3}\ln(1+x^2)-\frac{2}{9}x^3+\frac{2}{3}x-\frac{2}{3}\arctan x+C.$$

(3) $\int x\cos^2 x\,dx=\frac{1}{2}\int x(1+\cos 2x)dx=\frac{x^2}{4}+\frac{1}{4}\int x\,d(\sin 2x)$

$$=\frac{x^2}{4}+\frac{x}{4}\sin 2x-\frac{1}{4}\int\sin 2x\,dx=\frac{x^2}{4}+\frac{x}{4}\sin 2x+\frac{1}{8}\cos 2x+C.$$

(4) $\int x\tan x\sec^4 x\,dx=\int x\sec^3 x\,d(\sec x)=\frac{1}{4}\int x\,d(\sec^4 x)=\frac{x}{4}\sec^4 x-\frac{1}{4}\int\sec^4 x\,dx$

$$=\frac{x}{4}\sec^4 x-\frac{1}{4}\int(\tan^2 x+1)d(\tan x)$$

$$=\frac{x}{4}\sec^4 x-\frac{1}{12}\tan^3 x-\frac{1}{4}\tan x+C.$$

(5) $I=\int e^{2x}\cos 3x\,dx=\frac{1}{3}\int e^{2x}\,d(\sin 3x)=\frac{1}{3}e^{2x}\sin 3x-\frac{2}{3}\int e^{2x}\sin 3x\,dx$

$$=\frac{1}{3}e^{2x}\sin 3x+\frac{2}{9}\int e^{2x}\,d(\cos 3x)=\frac{1}{3}e^{2x}\sin 3x+\frac{2}{9}e^{2x}\cos 3x-\frac{4}{9}I,$$

则 $I=\frac{e^{2x}}{13}(3\sin 3x+2\cos 3x)+C.$

12. 解 (1) $\int x\arcsin x\,dx=\frac{1}{2}\int\arcsin x\,d(x^2)=\frac{x^2}{2}\arcsin x-\frac{1}{2}\int\frac{x^2}{\sqrt{1-x^2}}dx$

$$=\frac{x^2}{2}\arcsin x-\frac{1}{2}\int\frac{1-(1-x^2)}{\sqrt{1-x^2}}dx$$

$$=\frac{x^2}{2}\arcsin x-\frac{1}{2}\arcsin x+\frac{1}{2}\int\sqrt{1-x^2}\,dx$$

$$=\frac{x^2}{2}\arcsin x-\frac{1}{4}\arcsin x+\frac{x\sqrt{1-x^2}}{4}+C.$$

(2) $\int x^2\arctan x\,dx=\frac{1}{3}\int\arctan x\,d(x^3)=\frac{x^3}{3}\arctan x-\frac{1}{3}\int\frac{x^3}{1+x^2}dx$

$$=\frac{x^3}{3}\arctan x-\frac{1}{3}\int\left(x-\frac{x}{1+x^2}\right)dx=\frac{x^3}{3}\arctan x-\frac{x^2}{6}+\frac{1}{6}\ln(1+x^2)+C.$$

(3) $\int\arcsin x\arccos x\,dx=x\arcsin x\arccos x-\int\frac{x}{\sqrt{1-x^2}}(\arccos x-\arcsin x)dx$

$$=x\arcsin x\arccos x+\int(\arccos x-\arcsin x)d(\sqrt{1-x^2})$$

$$=x\arcsin x\arccos x+\sqrt{1-x^2}(\arccos x-\arcsin x)+2x+C.$$

13. 解 (1) 令 $\sqrt{x}=t$，则

$$\int\ln(1+\sqrt{x})dx=\int\ln(1+t)d(t^2)=t^2\ln(1+t)-\int\frac{t^2}{t+1}dt$$

$$=t^2\ln(1+t)-\int\left(t-1+\frac{1}{t+1}\right)dt$$

$$=t^2\ln(1+t)-\frac{t^2}{2}+t-\ln(1+t)+C$$

$$=(x-1)\ln(1+\sqrt{x})-\frac{x}{2}+\sqrt{x}+C.$$

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(2)$\int\frac{x\mathrm{e}^x}{\sqrt{\mathrm{e}^x-1}}\mathrm{d}x=2\int\frac{x\,\mathrm{d}(\mathrm{e}^x-1)}{2\sqrt{\mathrm{e}^x-1}}=2\int x\,\mathrm{d}(\sqrt{\mathrm{e}^x-1})=2x\sqrt{\mathrm{e}^x-1}-2\int\sqrt{\mathrm{e}^x-1}\,\mathrm{d}x$,其中

$$\int\sqrt{\mathrm{e}^x-1}\,\mathrm{d}x\xlongequal{\sqrt{\mathrm{e}^x-1}=t}\int t\cdot\frac{2t}{1+t^2}\mathrm{d}t=2\int\left(1-\frac{1}{1+t^2}\right)\mathrm{d}t=2t-2\arctan t+C$$

$$=2\sqrt{\mathrm{e}^x-1}-2\arctan\sqrt{\mathrm{e}^x-1}+C,$$

故$\int\frac{x\mathrm{e}^x}{\sqrt{\mathrm{e}^x-1}}\mathrm{d}x=(2x-4)\sqrt{\mathrm{e}^x-1}+4\arctan\sqrt{\mathrm{e}^x-1}+C.$

(3)$\int\frac{x\mathrm{e}^x}{(x+1)^2}\mathrm{d}x=\int\frac{[(x+1)-1]\mathrm{e}^x}{(x+1)^2}\mathrm{d}x=\int\frac{\mathrm{e}^x}{x+1}\mathrm{d}x-\int\frac{\mathrm{e}^x}{(x+1)^2}\mathrm{d}x$

$$=\int\frac{\mathrm{e}^x}{x+1}\mathrm{d}x+\int\mathrm{e}^x\,\mathrm{d}\left(\frac{1}{x+1}\right)=\int\frac{\mathrm{e}^x}{x+1}\mathrm{d}x+\frac{\mathrm{e}^x}{x+1}-\int\frac{\mathrm{e}^x}{x+1}\mathrm{d}x$$

$$=\frac{\mathrm{e}^x}{x+1}+C.$$

(4)$\int\frac{1+\sin x}{1+\cos x}\mathrm{e}^x\,\mathrm{d}x=\int\frac{1+2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\mathrm{e}^x\,\mathrm{d}x=\int\left(\frac{1}{2}\sec^2\frac{x}{2}+\tan\frac{x}{2}\right)\mathrm{e}^x\,\mathrm{d}x$

$$=\int\frac{1}{2}\sec^2\frac{x}{2}\cdot\mathrm{e}^x\,\mathrm{d}x+\int\mathrm{e}^x\tan\frac{x}{2}\mathrm{d}x=\int\mathrm{e}^x\,\mathrm{d}\left(\tan\frac{x}{2}\right)+\int\mathrm{e}^x\tan\frac{x}{2}\mathrm{d}x$$

$$=\mathrm{e}^x\tan\frac{x}{2}+C.$$

14. 解 由$\max\{1,x^2\}=\begin{cases}x^2, & x<-1,\\ 1, & -1\leqslant x\leqslant 1,\\ x^2, & x>1,\end{cases}$得$\int\max\{1,x^2\}\mathrm{d}x=\begin{cases}\frac{x^3}{3}+C_1, & x<-1,\\ x+C, & -1\leqslant x\leqslant 1,\\ \frac{x^3}{3}+C_2, & x>1.\end{cases}$

由原函数连续,得$\begin{cases}-\frac{1}{3}+C_1=-1+C,\\ \frac{1}{3}+C_2=1+C,\end{cases}$解得$C_1=C-\frac{2}{3},C_2=C+\frac{2}{3}$,则

$$\int\max\{1,x^2\}\mathrm{d}x=\begin{cases}\frac{x^3}{3}+C-\frac{2}{3}, & x<-1,\\ x+C, & -1\leqslant x\leqslant 1,\\ \frac{x^3}{3}+C+\frac{2}{3}, & x>1.\end{cases}$$

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